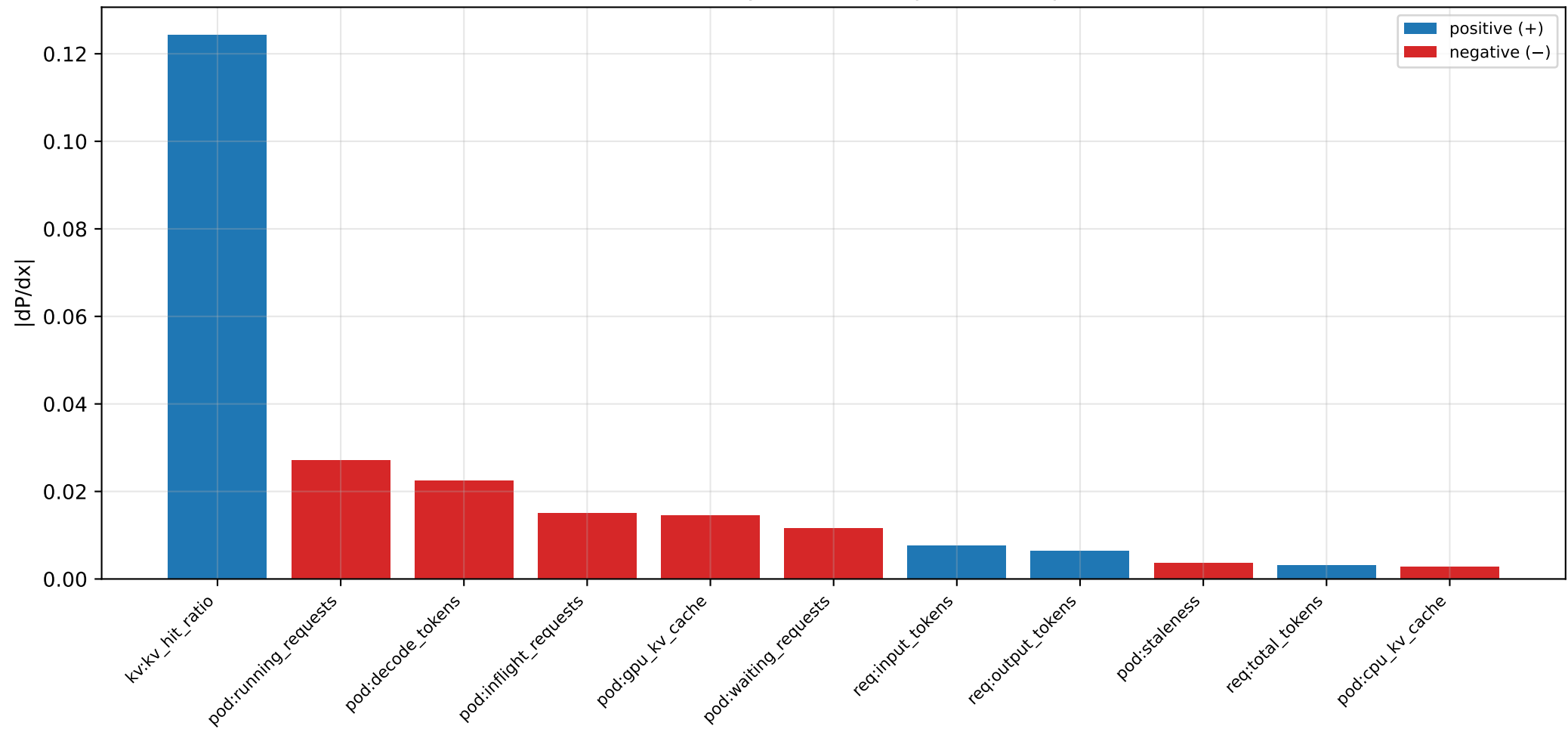


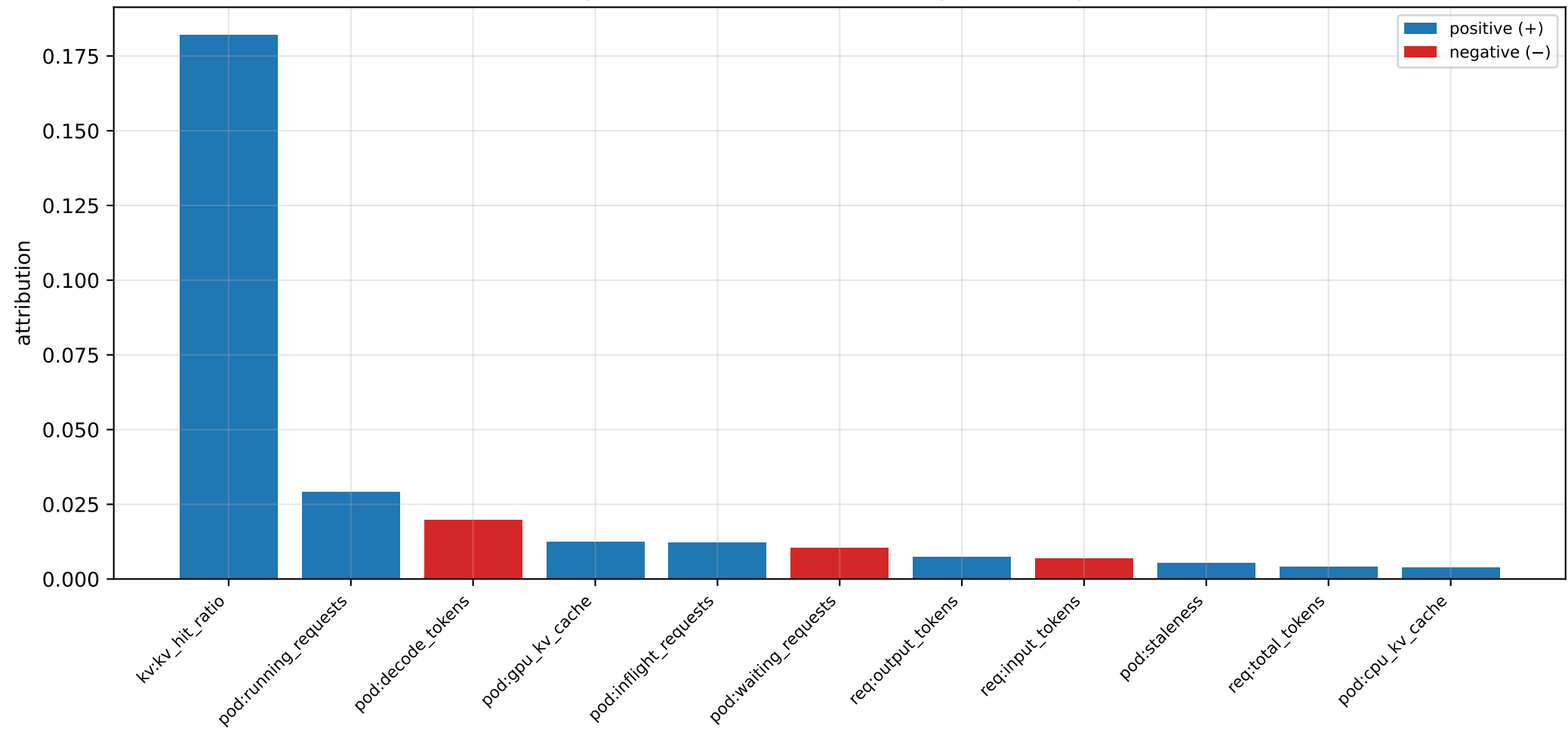
Routing Policy Explainability Report

final_model-data_replaced-processed-linear_simple_extended-lr_0.001-ttft_weight_0.5-ttftslo_2000-avgtpotslo_80-without_prefill_tokens-hidden_dim_128

Gradient Saliency (combined: pod, kv, request)



Integrated Gradients (combined: pod, kv, request)



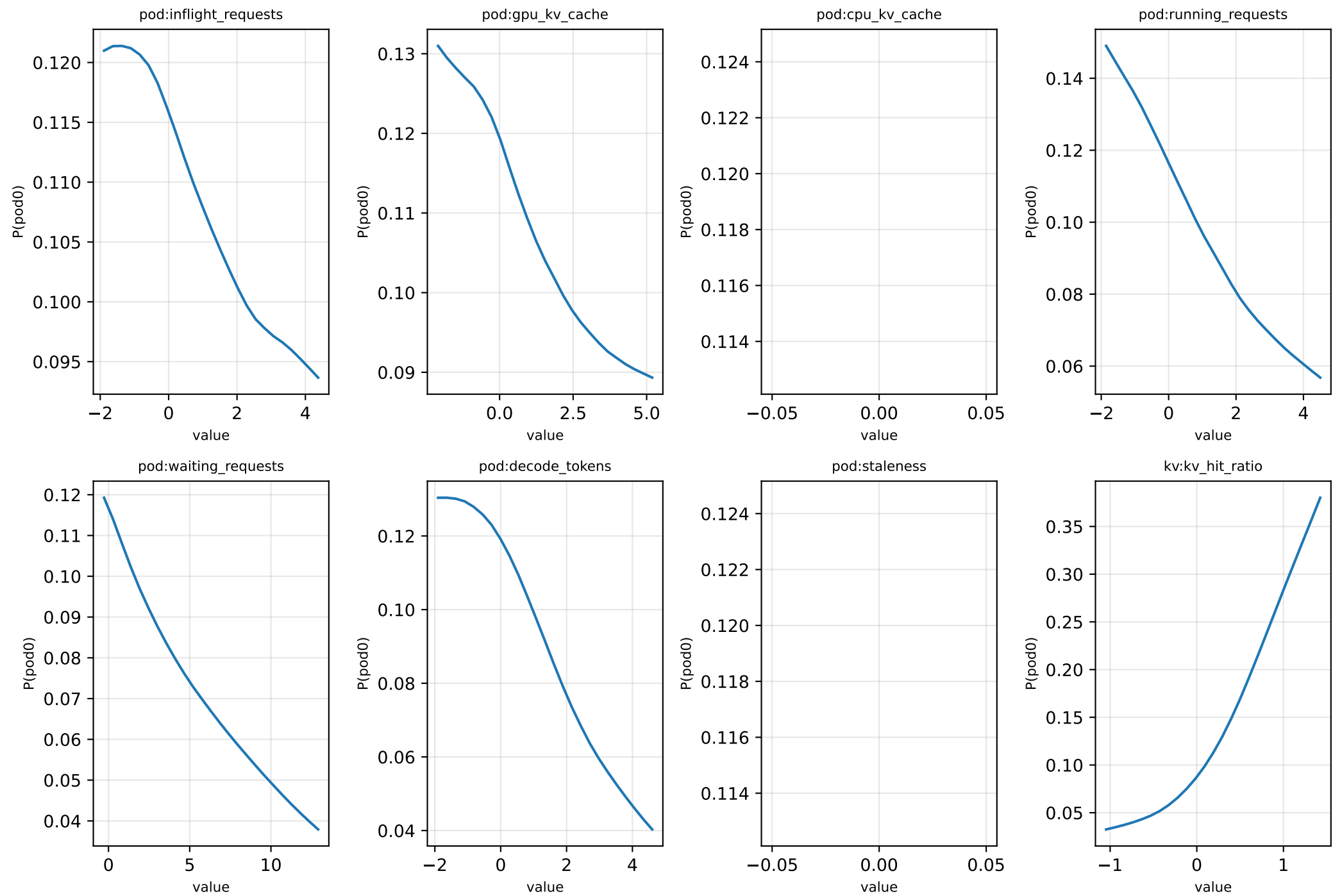
Permutation Symmetry Test

match_rate: 1.0

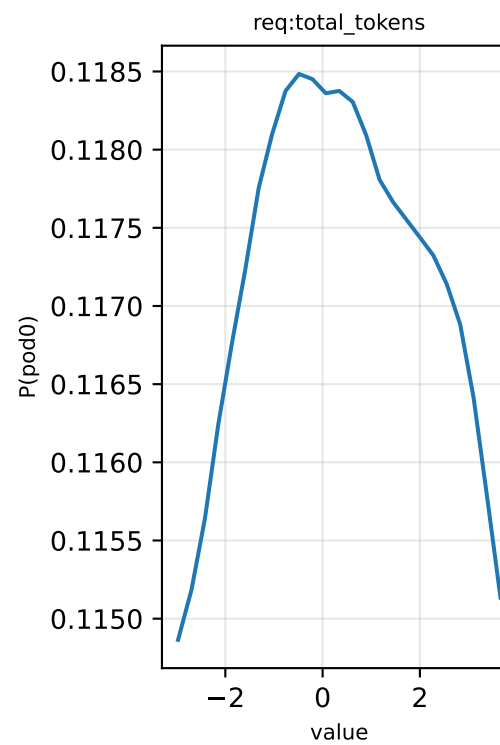
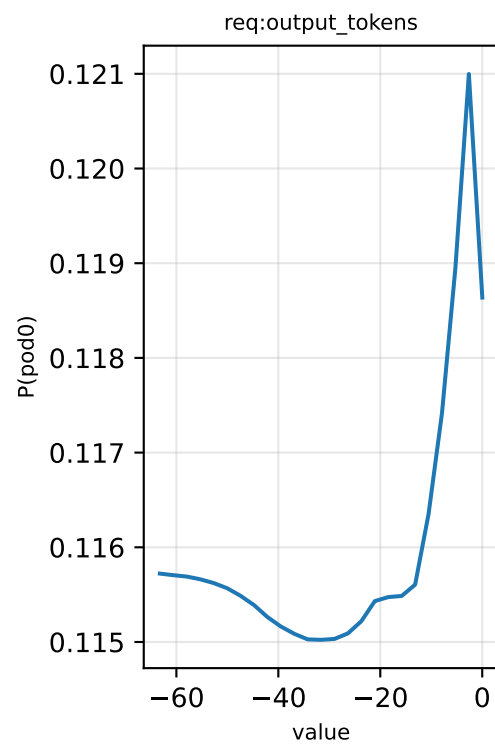
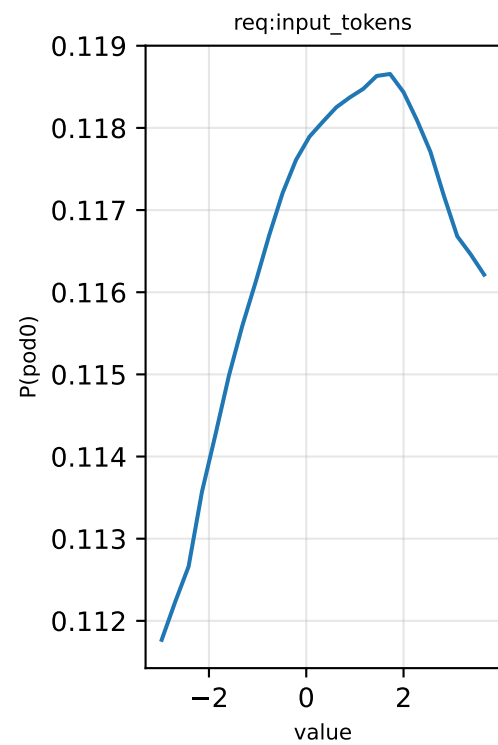
mean_l1_diff: 1.734920911008094e-08

Higher match_rate and lower mean_l1_diff indicate invariance to pod ordering.

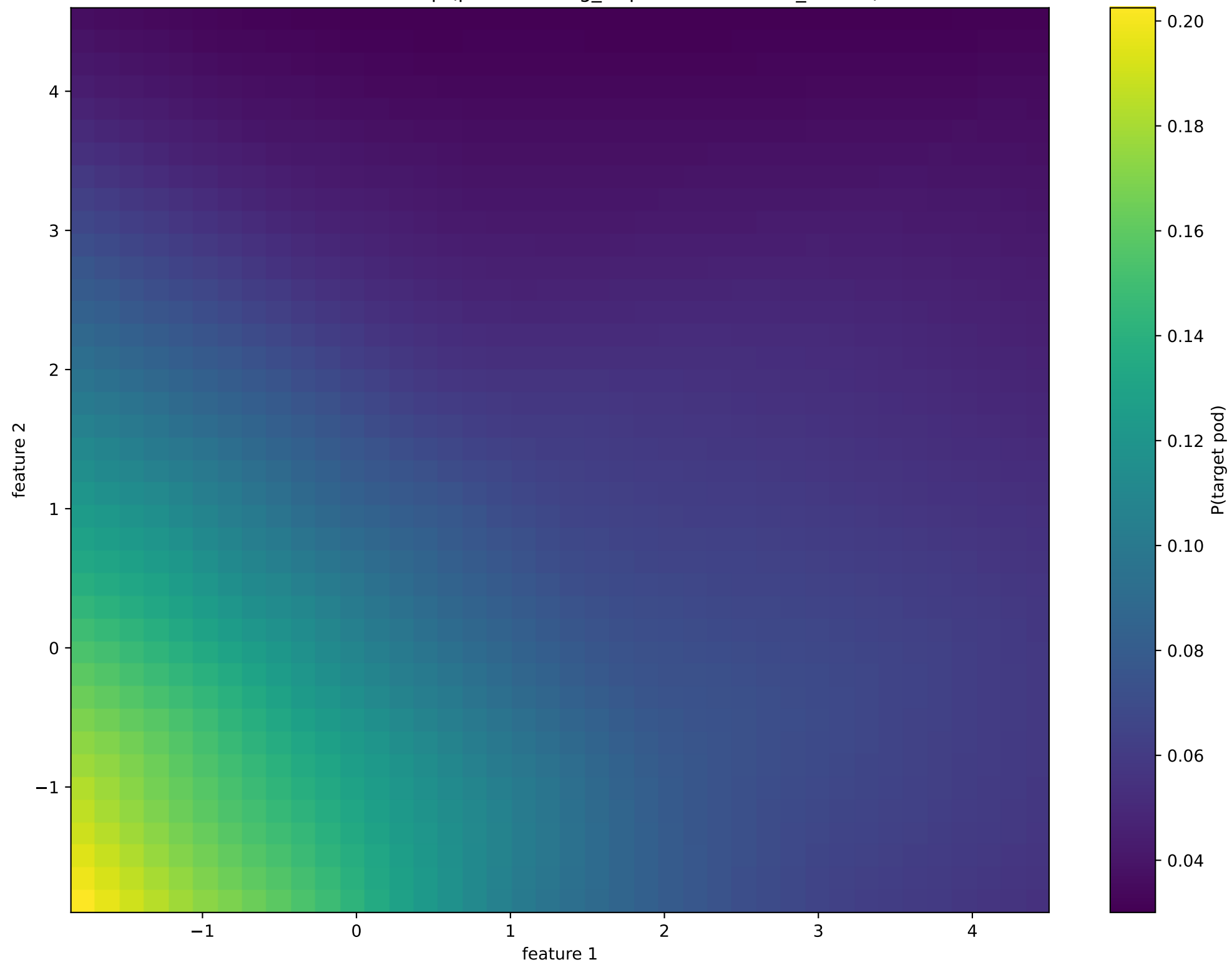
PDPs (all groups), target pod=0



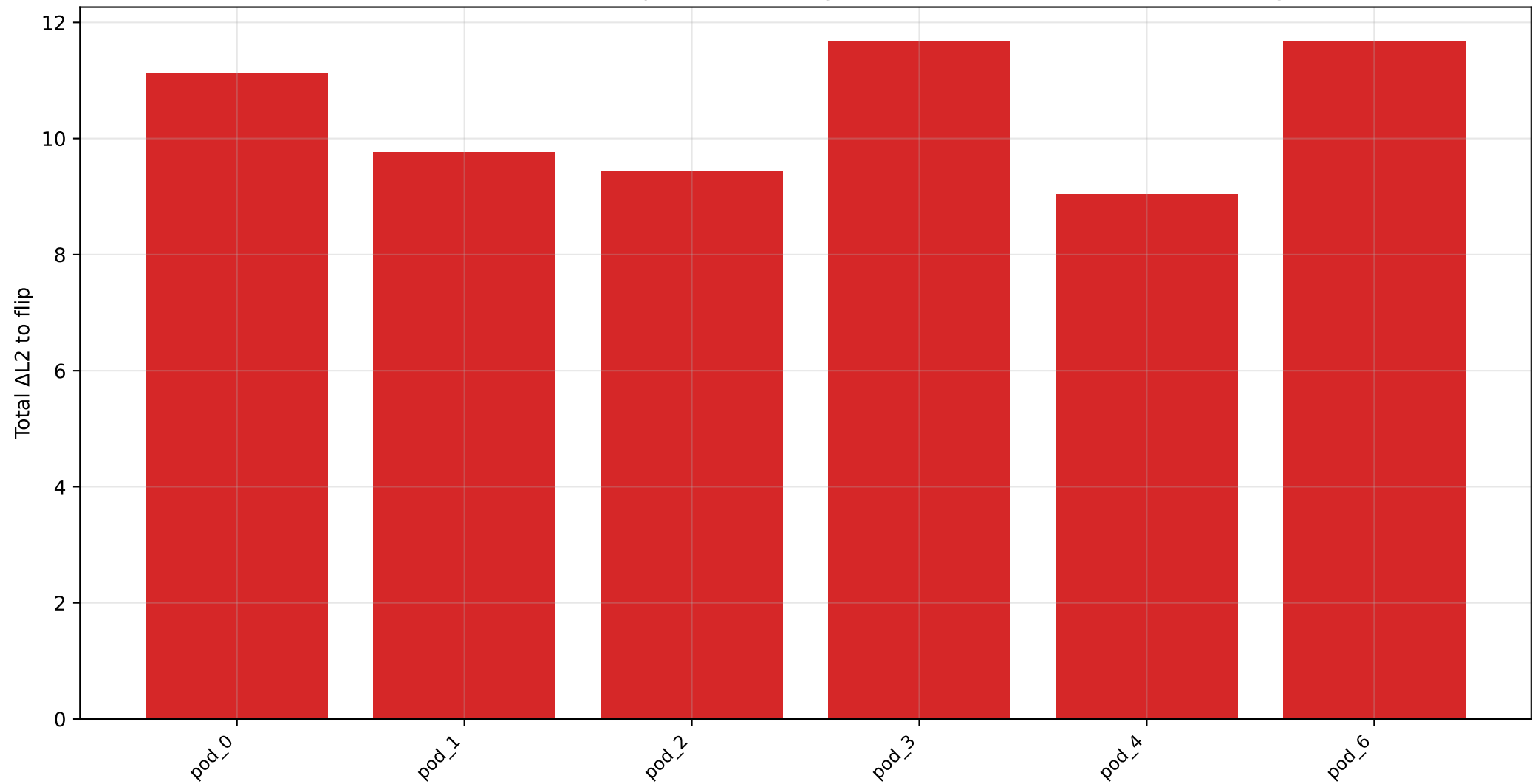
PDPs (all groups), target pod=0



Interaction heatmap (pod0 running_requests vs decode_tokens)



Counterfactual Distance to Flip to Each Pod (green=success, red=failed within budget)



EXPLAINABILITY FINDINGS

Saliency (local sensitivity):

- running_requests: 0.027
- decode_tokens: 0.023
- inflight_requests: 0.015
- gpu_kv_cache: 0.015
- waiting_requests: 0.012
- staleness: 0.004
- cpu_kv_cache: 0.003

Request Saliency:

- input_tokens: 0.008
- output_tokens: 0.006
- total_tokens: 0.003

KV Saliency:

- kv_hit_ratio: 0.124

Integrated Gradients (robust attribution):

- Top pod feature: running_requests
- KV importance: 0.182
- Request max attribution: output_tokens

PDP Trend Summary (target pod)

Pod features:

- inflight_requests: decreasing
- gpu_kv_cache: decreasing
- cpu_kv_cache: flat
- running_requests: decreasing
- waiting_requests: decreasing
- decode_tokens: decreasing
- staleness: flat

KV features:

- kv_hit_ratio: increasing

Request features:

- input_tokens: increasing
- output_tokens: increasing
- total_tokens: flat

Permutation Symmetry:

- match_rate: 1.0
- mean_l1_diff: 1.734920911008094e-08

Counterfactual Robustness:

- success_rate (flip achieved): 0.00
- min $\Delta L2$ to flip: 9.03
- mean $\Delta L2$ to flip: 10.45

Interpretation:

- High symmetry indicates no position bias.
- Larger $\Delta L2$ implies locally robust decisions (harder to flip).